

Network Port Scanning and Security Assessment

1. Objective

The objective of this task was to perform a network reconnaissance and security assessment on an authorized local network using Kali Linux. The assessment involved identifying active hosts, scanning for open TCP ports using Nmap, analyzing network packets with Wireshark, identifying potential security risks, and documenting the findings.

2. Tools Used

- Kali Linux
 - Nmap (Version 7.99)
 - Wireshark
 - VMware Workstation
 - Windows 11 (Host Operating System)
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3. Methodology

The following methodology was followed during the assessment:

1. Verified the installation of Nmap.
 2. Identified the local network configuration.
 3. Performed a TCP SYN scan using Nmap.
 4. Observed discovered hosts and scanned TCP ports.
 5. Captured network traffic using Wireshark during the Nmap scan.
 6. Analyzed TCP SYN and TCP Reset packets.
 7. Saved scan results for documentation.
 8. Documented the findings and security observations.
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4. Commands Used

Verify Nmap Installation

```
nmap --version
```

Display Network Configuration

```
ip addr
```

Perform TCP SYN Scan

```
sudo nmap -sS <masked-network>/24
```

Save Scan Results

```
sudo nmap -sS <masked-network>/24 -oN Scan_Result.txt
```

```
sudo nmap -sS <masked-network>/24 -oX Scan_Result.xml
```

5. Findings

- The TCP SYN scan was successfully executed on the authorized network.
 - The target host was reachable.
 - The scan did not identify any open TCP ports on the scanned host.
 - The scan indicated that the scanned ports were filtered or blocked by firewall rules.
 - Wireshark successfully captured the TCP SYN packets generated by the Nmap scan.
 - TCP RST/ACK responses were observed, indicating that the scanned ports were closed.
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6. Security Risks

The assessment identified the following observations:

- No open TCP ports were exposed during the scan.
 - Filtered ports indicate that firewall protection is active.
 - Proper firewall configuration reduces the network attack surface.
 - If unnecessary ports are opened in the future, they may become potential entry points for attackers.
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7. Recommendations

- Continue using firewall protection to restrict unauthorized access.
 - Regularly perform network scans to detect unintended exposed services.
 - Keep operating systems and network services updated with the latest security patches.
 - Disable unnecessary network services.
 - Monitor network traffic periodically using Wireshark or other monitoring tools.
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8. Conclusion

The network port scanning exercise was successfully completed using Kali Linux, Nmap, and Wireshark. The assessment demonstrated the process of discovering network hosts, analyzing TCP SYN scan traffic, and evaluating the security posture of an authorized network. The absence of open TCP ports and the presence of filtered ports indicate that the target system was appropriately protected against unauthorized network access.